Within the flat 6D Conformal Spacetime Algebra system ($Cl(4,1,1)$), a Complementary Metal-Oxide-Semiconductor (CMOS) inverter can be modeled as a coupled electro-gravitational system.

Instead of treating the gate field-effect as a phenomenological quantum statistical parameter, this model derives channel conduction directly from the inhomogeneous master field equation $D \cdot F_{\text{total}} = J_{\text{total}}$ [2.4] acting on mobile electron solitons [4.1] and localized core vacancies (holes) [5.5].

Section 5.7: The CMOS Inverter (Complementary Soliton Logic)

A CMOS inverter consists of an NMOS transistor (pull-down network) and a PMOS transistor (pull-up network) connected in series between the supply potential V_{DD} and the ground potential GND. The input potential V_{in} is connected to both gates, and the output potential V_{out} is taken from the common drain connection.

```
           V_DD
            |
        +---+---+
        |  PMOS | (Pull-Up Network:
  V_in --+   o---+   Conduction by Positive Vacancies)
        |       |
        +---+---+
            |
            +------ V_out (Output Node)
            |
        +---+---+
        |  NMOS | (Pull-Down Network:
  V_in --+       +   Conduction by Electron Solitons)
        |       |
        +---+---+
            |
           GND
```

5.7.1 The Static Field-Effect Channel Equation The gate terminal is separated from the semiconductor channel by a thin silicon dioxide (SiO_2) dielectric layer of thickness t_{ox}. Applying an input potential V_{in} to the gate generates a perpendicular electric field E_{gate} across the oxide.

In the static limit, the inhomogeneous master field equation $D \cdot F_{\text{total}} = J_{\text{total}}$ simplifies to the Gauss relation [2.4]:

$$\nabla \cdot E_{\text{gate}} = \frac{\rho_e}{\epsilon_{\text{ox}}}$$

This vertical electric field polarizes the underlying semiconductor channel, attracting or repelling charge carriers depending on the substrate doping.

5.7.2 NMOS Channel Conduction (Electron Soliton Drift) The NMOS transistor is fabricated on a p-type substrate. * **Inversion Layer Formation:** When V_{in} exceeds a threshold potential $V_{th,n}$, the vertical electric field $E_{\text{gate, N}}$ repels localized lattice vacancies [5.5] and attracts mobile electron solitons (loops [4.1]) to the surface. * **Soliton Surface Density:** The accumulated surface charge density $\mathcal{Q}_n(y)$ along the channel length y is:

$$\mathcal{Q}_n(y) = C_{\text{ox}} \left[V_{in} - V_{th,n} - V(y) \right]$$

where: * $C_{\text{ox}} = \frac{\epsilon_{\text{ox}}}{t_{\text{ox}}}$ is the oxide capacitance per unit area. * $V(y)$ is the local channel potential at distance y from the source. * **Current Flow ($I_{ds,n}$):** Under a lateral drain-to-source electric field $E_{\text{lateral}} = -\frac{dV}{dy}$, the electron solitons drift toward the drain with velocity $u_{\text{drift},n} = \mu_n \frac{dV}{dy}$, where μ_n is the soliton mobility in the silicon lattice. Integrating over the gate width W and length L yields:

$$I_{ds,n} = \frac{W}{L} \mu_n C_{\text{ox}} \left[(V_{in} - V_{th,n}) V_{\text{out}} - \frac{1}{2} V_{\text{out}}^2 \right] \quad \text{(for linear operation)}$$

5.7.3 PMOS Channel Conduction (Positive Vacancy Drift) The PMOS transistor is fabricated on an n-type substrate. * **Inversion Layer Formation:** When V_{in} falls below $V_{DD} - |V_{th,p}|$, the vertical electric field $E_{\text{gate, P}}$ repels mobile electron solitons and attracts positive core vacancies (holes [5.5]) to the surface. * **Vacancy Surface Density:** The accumulated surface charge density $\mathcal{Q}_p(y)$ along the channel is:

$$\mathcal{Q}_p(y) = C_{\text{ox}} \left[V_{DD} - V_{in} - |V_{th,p}| - (V_{DD} - V(y)) \right]$$

* **Current Flow ($I_{sd,p}$):** The positive vacancies drift under the lateral electric field with mobility μ_p. Integrating along the channel length yields:

$$I_{sd,p} = \frac{W}{L} \mu_p C_{\text{ox}} \left[(V_{DD} - V_{in} - |V_{th,p}|)(V_{DD} - V_{\text{out}}) - \frac{1}{2} (V_{DD} - V_{\text{out}})^2 \right] \quad \text{(for linear operation)}$$

5.7.4 Inverter Switching Dynamics The output node is characterized by a lumped load capacitance C_{load} (representing the input capacitance of subsequent stages and parasitic routing wires). The charge conservation equation ($\nabla \cdot J = -\frac{\partial \rho}{\partial t}$) at the output node dictates:

$$C_{load} \frac{dV_{out}}{dt} = I_{sd,p} - I_{ds,n}$$

This differential equation governs the transition between the two stable logical states:

State A: Input LOW ($V_{in} \to 0$)

1. **NMOS State:** Since $V_{in} < V_{th,n}$, the vertical electric field is insufficient to attract electron solitons. The channel density $\mathcal{Q}_n \approx 0$, rendering the NMOS channel non-conductive:

$$I_{ds,n} \approx 0$$

2. **PMOS State:** Since $V_{in} \ll V_{DD} - |V_{th,p}|$, the PMOS gate electric field is highly attractive to positive core vacancies. The channel is conductive:

$$I_{sd,p} > 0$$

3. **Output Transition:** The output capacitance charges through the PMOS channel until the output potential rises to the supply potential:

$$V_{out} = V_{DD}$$

State B: Input HIGH ($V_{in} \to V_{DD}$)

1. **PMOS State:** Since $V_{in} > V_{DD} - |V_{th,p}|$, the gate electric field repels positive vacancies. The channel density $\mathcal{Q}_p \approx 0$, rendering the PMOS channel non-conductive:

$$I_{sd,p} \approx 0$$

2. **NMOS State:** Since $V_{in} \gg V_{th,n}$, the gate electric field is highly attractive to mobile electron solitons. The channel is conductive:

$$I_{ds,n} > 0$$

3. **Output Transition:** The output capacitance discharges through the NMOS channel to ground, pulling the output potential down:

$$V_{out} = 0$$

5.7.5 Power Dissipation and Short-Circuit Current In both static states (State A and State B), one of the complementary channels is non-conductive, meaning the steady-state current from V_{DD} to ground is zero:

$$I_{\text{static}} = 0$$

During the switching transition, however, both V_{in} passes through the intermediate region ($V_{th,n} < V_{in} < V_{DD} - |V_{th,p}|$). During this brief interval, both channels are partially conductive, allowing a short-circuit current of electron solitons to flow directly from the supply to ground and recombine with positive vacancies at the drain junction. This transition current drives the dynamic power dissipation of the CMOS system:

$$P_{\text{dynamic}} = C_{load} V_{DD}^2 f$$

where f is the switching frequency. This completes the CMOS inverter model, describing the complementary logic gates through the controlled drift of positive and negative soliton distributions.